

UNIT 4 :Simplification of Boolean Algebra

Presented by

Er.Harendra Bikram Shah
(Lecturer)

OUTLINES

- ❖ K-Map of 2,3,4 variables
- ❖ Simplification and Realization using NAND and NOR gates
- ❖ Practical Design Steps



K-Map

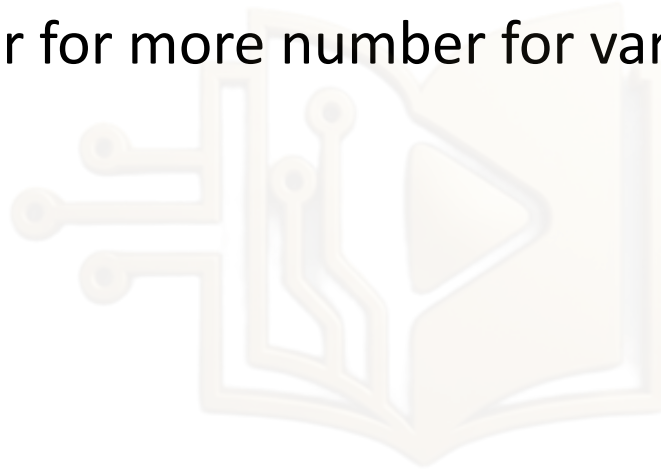
Introduction to K-map:

- In previous chapters, we have simplified the Boolean functions using Boolean postulates and theorems.
- It is a time consuming process and we have to re-write the simplified expressions after each step.
- To overcome this difficulty, Karnaugh introduced a method for simplification of Boolean functions in an easy way.
- **A Karnaugh map (K-map) is a pictorial method used to minimize Boolean expressions without having to use Boolean algebra theorems and equation manipulations.**
- A K-map can be thought of as a special version of a truth table .
- It is a graphical method, which consists of 2^n cells for 'n' variables.
- In 1953, the American Physicist Maurice **Karnaugh** invented K-Map

K-Map

Significance/Application/Importance of K-map:

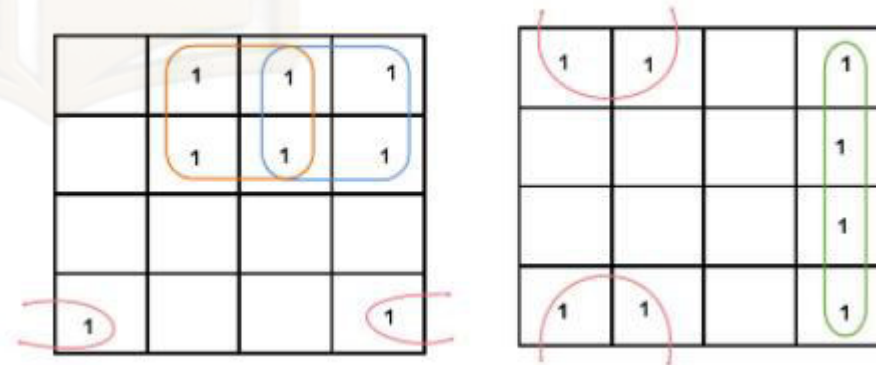
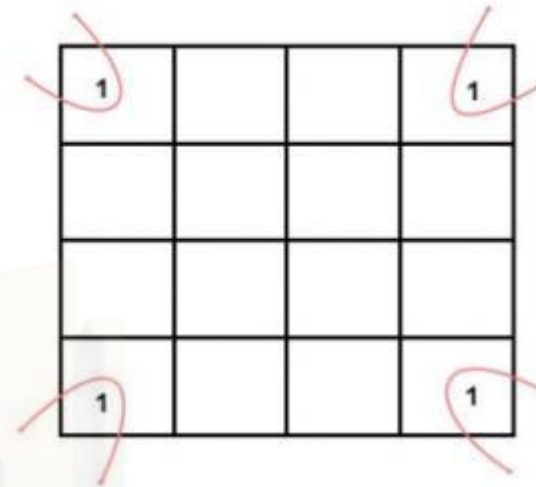
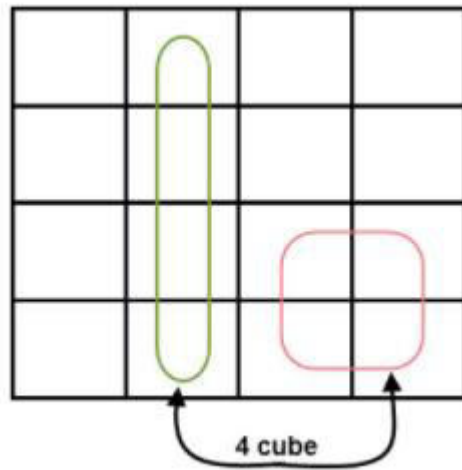
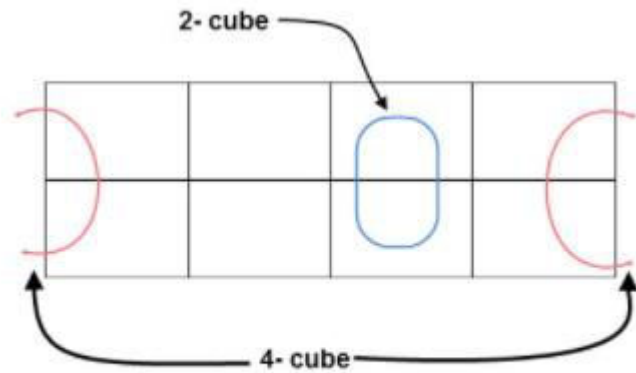
- Boolean algebra can be simplified systematically
- K-map reduces logic functions quickly and simply.
- K-maps are both faster and Easier for more number for variables like 4 or more numbers of variables.



Steps to solve expression using K-map-

1. Select K-map according to the number of variables.
2. Identify minterms or maxterms as given in problem.
3. For SOP put 1's in blocks of K-map respective to the minterms (0's elsewhere).
4. For POS put 0's in blocks of K-map respective to the maxterms (1's elsewhere).
5. Make rectangular groups containing total terms in power of two like 2, 4, 8 .. (except 1) and try to cover as many elements as you can in one group.
6. Don't care "x" should also be included while grouping to make a larger possible group.
7. From the groups made in step 5 find the product terms and sum them up for SOP form.

Pairing Ways in k-map



SOP FORM

K-Map of 2 variables

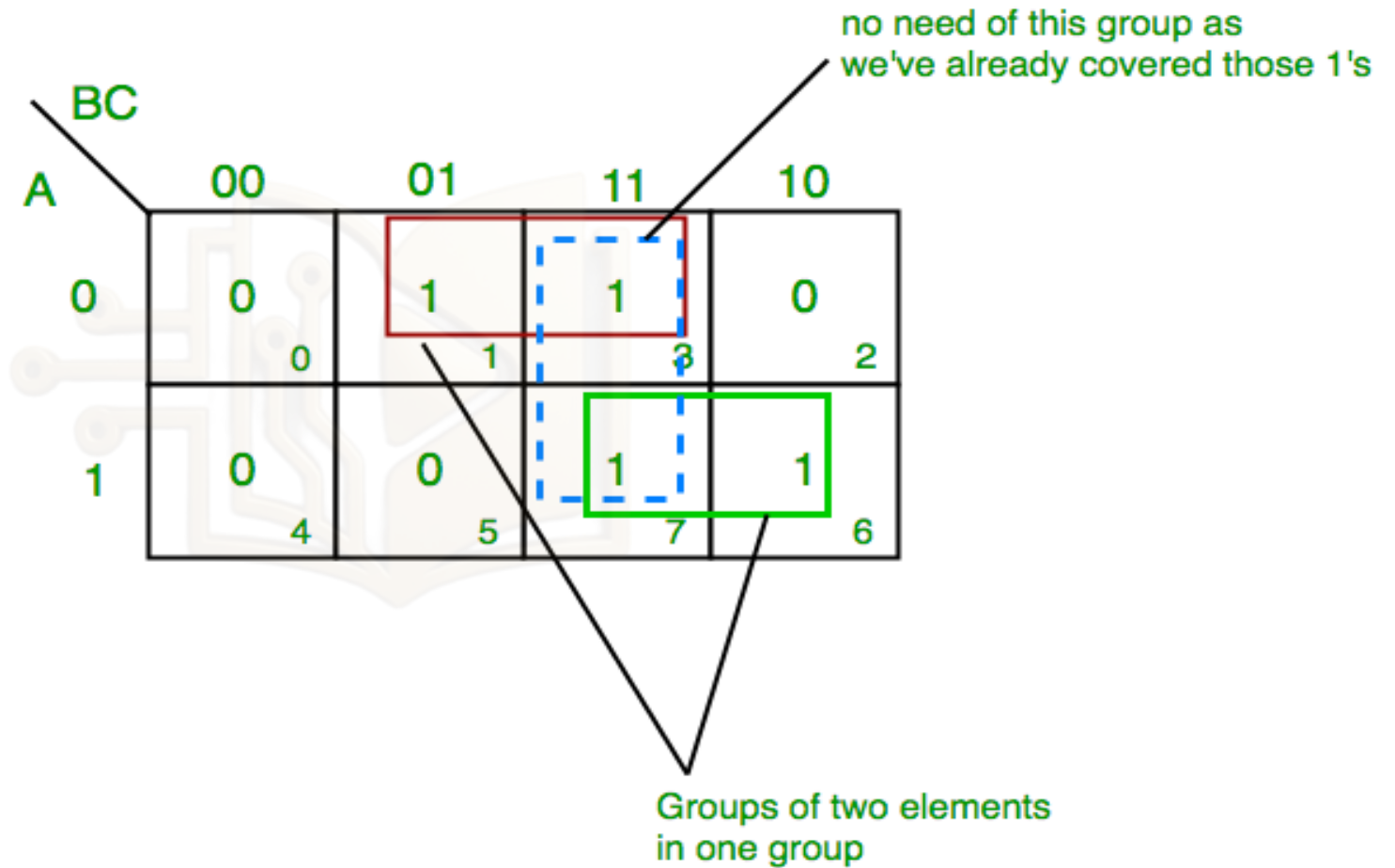
		B	
		0	1
A	0	$A'B'$ 0	$A'B$ 1
	1	AB' 2	AB 3

K-Map of 3 variables

SOP FORM

1.K-map of 3 variables-

$$Z = \sum A, B, C(1, 3, 6, 7)$$



From **red** group we get product term—

$A'C$

From **green** group we get product term

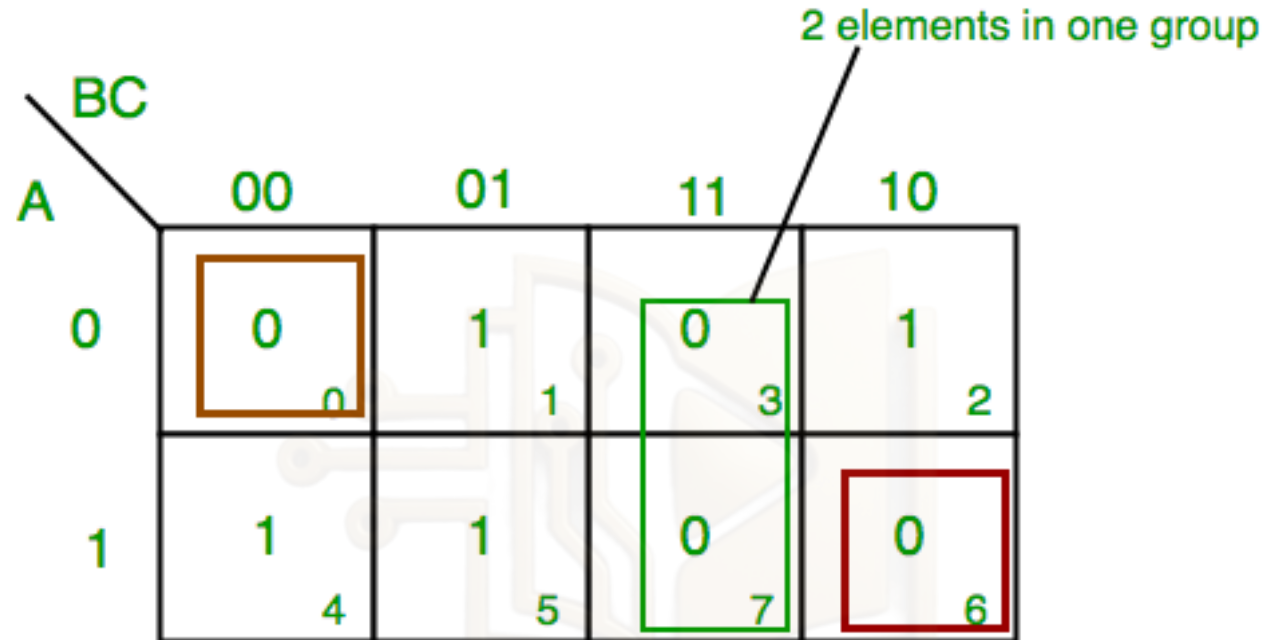
AB

Summing these product terms we get- **Final expression**

$(A'C + AB)$

SoP FORM

K-Map of 3 variables



SoP FORM

K-Map of 4 variables

		CD			
		$C'D'$	$C'D$	CD	CD'
A B	$A'B'$ 00	0	1	3	2
	$A'B$ 01	4	5	7	6
AB	11	12	13	15	14
	AB' 10	8	9	11	10

PoS FORM

K-Map of 4 variables

		C+D			
		C+D		C'+D'	
		C+D'		C'+D	
A+B					
A+B 0+0		0	1	3	2
A+B' 0+1		4	5	7	6
A'B' 1+1		12	13	15	14
A'+B 1+0		8	9	11	10

5 Variable K-Map

The number of cells in 5 variable K-map is thirty-two, since the number of variables is 5. The following figure shows 5 variable K-Map.

- There is only one possibility of grouping 32 adjacent min terms.
- There are two possibilities of grouping 16 adjacent min terms. i.e.,
- grouping of min terms from m_0 to m_{15} and m_{16} to m_{31} .
- If $v=0$, then 5 variable K-map becomes 4 variable K-map.

V=0

YZ

WX

00

01

11

10

00

01

11

10

m₀

m₁

m₃

m₂

m₄

m₅

m₇

m₆

m₁₂

m₁₃

m₁₅

m₁₄

m₈

m₉

m₁₁

m₁₀

V=1

YZ

WX

00

01

11

10

00

01

11

10

m₁₆

m₁₇

m₁₉

m₁₈

m₂₀

m₂₁

m₂₃

m₂₂

m₂₈

m₂₉

m₃₁

m₃₀

m₂₄

m₂₅

m₂₇

m₂₆

In the above all K-maps, we used exclusively the min terms notation. Similarly, you can use exclusively the Max terms notation.

K-Map with Don't care

- Everything is same except the notation 'X' is used for don't care condition
- Don't care condition will remain same for both POS and SOP form

